

Image-Informed Urban Propagation and Route-Level Body Exposure Across Ten Cities at 15 GHz

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ABSTRACT Street-level radiofrequency exposure varies along a pedestrian route because buildings change the direct and reflected fields, surface materials affect how much power returns to the street, and the human body absorbs differently depending on the direction of arrival. This study projects 360-degree street images onto a photogrammetric city mesh to identify surface materials along pedestrian routes at 15 GHz in ten cities across Europe, Latin America, and East Asia. Transmitters are placed along the visible roofline in proportion to its physical length. All results are normalized per unit product of areal source density and effective isotropic radiated power (EIRP). The transport model treats direct paths and one specular reflection exactly, estimates one diffuse reflection, and stops. The arriving fields and their directions are then applied to a 56,024-element Duke body model. The ten routes contain 163 observation points, each calculated with 64 independent replicas of 200,000 primary rays and 4,096 angular output cells. In a controlled test scene, the adjoint estimate and deterministic quadrature differ by at most 0.0616 dB for the reflected term. An independent forward tracer gives a maximum difference of 0.0344 dB. Route-median whole-body specific absorption rate differs by a factor of 14.31 across the ten routes. Direct paths carry the largest share at 156 of the 157 points with line of sight, while the diffuse reflection is the only nonzero modeled contribution at six fully shadowed points. The maximum change from 48 to 64 replicas is 0.0139 dB, though lower-tail estimates in Mexico City and Tokyo are less stable. These results describe ten selected routes under a fixed model and do not estimate city-wide or deployed-network exposure.

INDEX TERMS Body dosimetry, image-informed propagation, ray tracing, street imagery, urban propagation, whole-body specific absorption rate.

I. INTRODUCTION

AS wireless networks expand into higher frequency bands, the built environment plays a larger role in determining the radiofrequency field that reaches a pedestrian [1]. Along a city street, received power can change over a few meters. A person walking past a row of buildings can move from a clear view of a rooftop transmitter into a shadow where buildings block the direct field and only reflected power reaches the street. The facade materials then determine how much power returns to the street [1], [2]. This variation also matters after the field arrives: whole-body absorption depends on the direction of arrival and on how the body is oriented, so a single power value at a single point does not describe exposure along a walking route. A route-level calculation must keep the arrival direction until the field is applied to the body [3].

Ray tracing computes detailed propagation paths between transmitters and receivers in a three-dimensional city model [4]. It has been used for city-scale downlink exposure using published base-station locations [5], and to compute propagation and body exposure along an outdoor pedestrian path [6]. Stochastic geometry models describe exposure when individual transmitter locations are not known [7], while pre-computed body coefficients separate the incident field from the absorption calculation [8]. Together, these studies cover deterministic urban propagation, spatial source models, and the directional absorption step that converts the arriving field into absorbed power on the body. Because they use different source assumptions, their reported exposure values are not directly comparable without a common source normalization.

A photogrammetric city mesh gives accurate building geometry, but its triangles carry no material information.

Street-level images can fill that gap. The Mapillary Vistas dataset provides a taxonomy for dense segmentation of street scenes [9]. Kamari *et al.* segment street-level images, project the resulting material classes onto city geometry, and use that geometry in millimeter-wave ray tracing [10]. Xia *et al.* use semantic point-cloud classification and detailed scene reconstruction for outdoor ray tracing at 2.8 GHz [11]. Projecting image-derived materials onto city geometry is therefore established. This study does not claim any of these operations as new. It uses them to build a material map around fixed pedestrian routes, keeping the default geometry-based material wherever the images give no reliable label.

This study combines these established parts in one fixed model. Each 360-degree street image is aligned with the same photogrammetric city mesh used for ray tracing, and the image labels are projected onto that mesh to produce a material map. A common transmitter model distributes sources along the visible roofline at every site. Rooflines are a natural choice: they are elevated, street-facing, and visible from the route. Distributing transmitters in proportion to roofline length avoids tying the result to any particular deployment. The ray tracer keeps the direct, specular, and diffuse components separate, preserving their arrival directions, until the field is applied to the body. The transport model treats direct paths and one specular reflection exactly, estimates one diffuse reflection, and stops there. The route points are fixed case studies, not a population sample, and the roofline transmitters are a model, not a measured deployment. To the best of the authors' knowledge, prior work has not combined aligned 360-degree street images, a common roofline transmitter model, and separate transport components with their arrival directions in a single route-level body exposure calculation.

This study applies the method at 15 GHz along pedestrian routes in ten cities: Ghent, Prague, Madrid, Mexico City, Tokyo, Brussels, London, Milan, Krakow, and Toulouse. The ten routes contain 163 observation points with image-derived surface materials and a body model facing along each walk. Every result is normalized per unit active-source areal density and per unit effective isotropic radiated power (EIRP). The route-median whole-body SAR values differ by a factor of 14.31 across the ten routes. These results describe fixed routes under one source and transport model. They do not estimate population exposure or deployed-network exposure.

This work makes the following three contributions.

- 1) Aligned 360-degree street images supply traceable material labels to the same city mesh used for ray tracing. Surfaces without a reliable image label keep their geometry-based material.
- 2) A roofline transmitter model and transport calculation keep direct, specular, and diffuse power separate, with arrival directions, until the field is applied to the body.
- 3) An application across ten urban routes reports fixed-route exposure distributions, controlled first-diffuse validation, replica convergence, and the retained transport at the six observation points where buildings block

TABLE 1. Ten fixed routes with observation-point counts, route spans, and computation times

Site	Route points	Route span (m)	Wall time (s)
Brussels	14	87.00	32.16
Ghent	10	49.04	6.59
Krakow	16	116.10	31.76
London	22	121.10	73.77
Madrid	14	73.47	8.06
Mexico City	11	60.79	5.95
Milan	23	128.43	69.02
Prague	22	119.39	39.00
Tokyo Hachiko	16	87.38	10.02
Toulouse	15	82.47	37.87
Total	163	925.17	314.20

all direct and specular paths.

II. METHOD

A. STUDY CONFIGURATION

Fig. 1 shows the study configuration. A 360-degree street image is aligned with a photogrammetric city mesh cropped to a 250 m radius around the route. Object and material labels from the image are projected onto the visible mesh triangles without changing their geometry. The roofline visible from the route gives the possible transmitter locations. Fixed points along the pedestrian route give the observation positions, and the anatomical body model faces along the direction of travel at each point.

Table 1 lists the ten selected routes. Each route follows a connected street corridor covered by aligned 360-degree street images. The calculation places fixed observation points along that corridor, including interpolated positions between image locations, so the number of images and the number of observation points can differ. The 163 route points are fixed observations, not a random sample of pedestrians or places. The body model faces along the direction of travel, so a different route would change both position and orientation.

Two image-analysis models work in sequence. Mask2Former assigns a Vistas object class (building, road, vegetation, etc.) to every image pixel [9], [12]. SAM 3 then tests material and vegetation labels inside the compatible object regions [13]. Both sets of labels are projected onto the visible city mesh using the known camera position and viewing direction. Where multiple images cover the same triangle, the labels are combined into one material map, and every mapped triangle stays linked to its source image. A mesh triangle changes material only when the object and material labels agree and pass the acceptance tests. All other triangles keep their default material. The prompts, alignment tests, rejected labels, and mapping rules are given in the supplementary material.

Fig. 2 shows the computation and its input checks. A hash-verified manifest lists every input file: aligned images, material map, city mesh, route, roofline, body model, and transport settings. Only files whose hashes match the manifest enter the ten-site data set. The transport calculation keeps direct, specular, and diffuse contributions separate until the field is

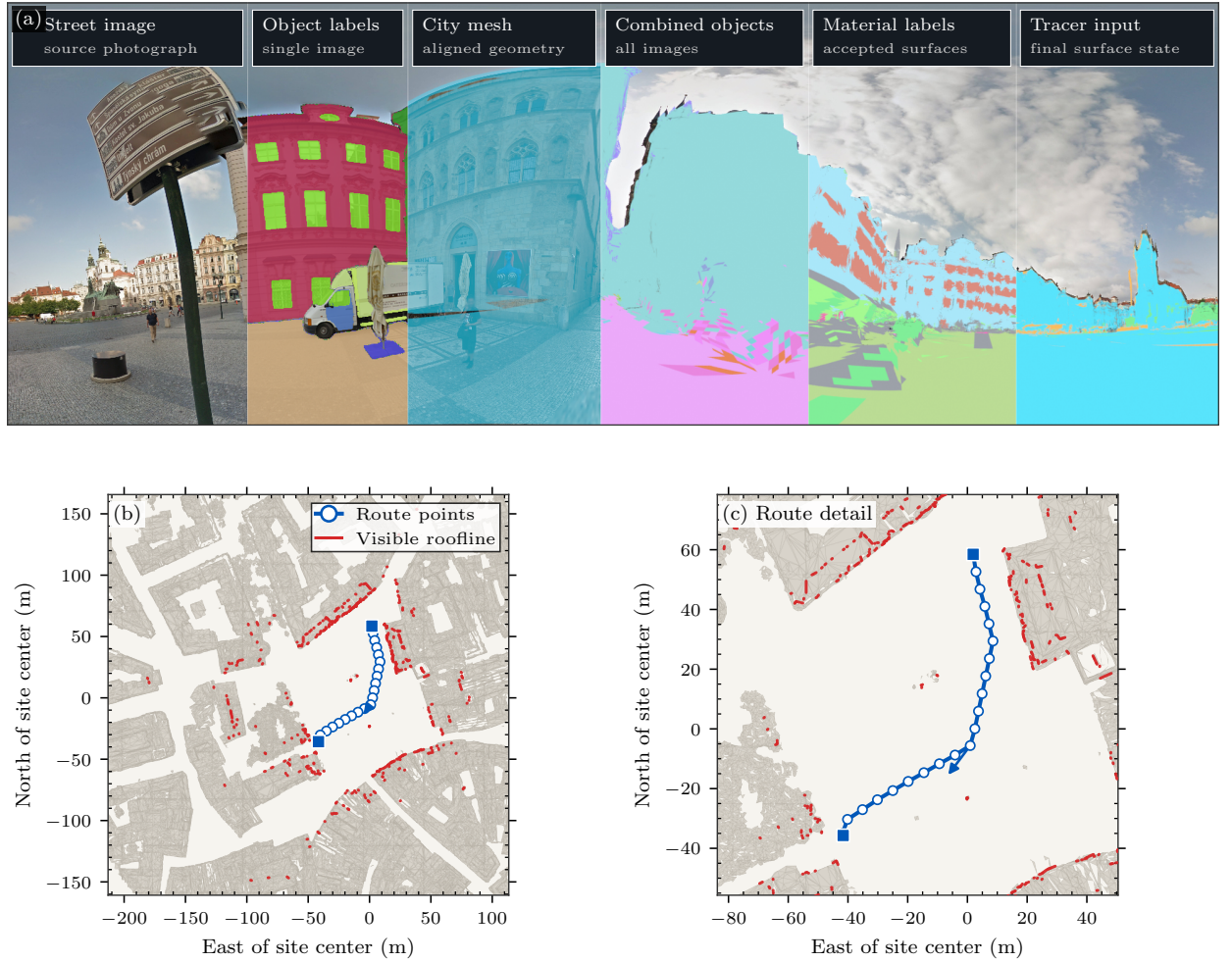


FIGURE 1. Study configuration at Prague Old Town Square. (a) A 360-degree street image is segmented, projected onto the city mesh, and converted to the material map used for ray tracing. (b) The 22 observation points and the visible roofline on a plan view. (c) Close-up of the route. The arrow shows the body model's direction of travel.

applied to the body. The ten manifests contain 420 verified entries. Across the resulting 10,432 directional body fields, the largest residual when the three components are summed back to the stored total is $1.214 \times 10^{-17} \text{ m}^{-2}$.

B. EXPOSURE CALCULATION

The calculation uses a fixed route, city mesh, material map, and roofline transmitter model. Every route point uses 15 GHz and a crop with radius $R_{\text{crop}} = 250 \text{ m}$, which gives $A_{\text{crop}} = \pi R_{\text{crop}}^2 = 196,349.54 \text{ m}^2$. The receiver position $\mathbf{x}$ is the ray origin and body reference point. The body model faces along the local direction of travel. Index i denotes a roofline segment, $r_i(\mathbf{x})$ is its range to the receiver, and $\mathbf{r}$ is a position on the body surface. These assumptions are fixed across replicas and sites.

The exact positions of future transmitters are unknown, but elevated rooflines are plausible street-facing locations. The model therefore distributes the expected transmitters uni-

formly per unit physical length of the roofline visible from the route. Let ρ_A be the expected active-site density and let A_{crop} be the crop area. For a roofline segment with endpoints $\mathbf{p}_i$ and $\mathbf{p}_{i+1}$, ℓ_i is its physical three-dimensional length. The expected number of transmitters and the fraction assigned to segment i are

$$N_{\text{site}} = \rho_A A_{\text{crop}}, \quad p_i = \frac{\ell_i}{\sum_j \ell_j}, \quad \ell_i = \|\mathbf{p}_{i+1} - \mathbf{p}_i\|_2. \quad (1)$$

Thus, N_{site} sets the number of transmitters and p_i assigns a fraction of that number to segment i . Dividing by the total roofline length ensures that splitting one segment into smaller numerical pieces does not add transmitters. The baseline uses three-dimensional length rather than horizontal projected length.

An unobstructed reference separates the source density from the scene geometry. The reference sums the inverse-

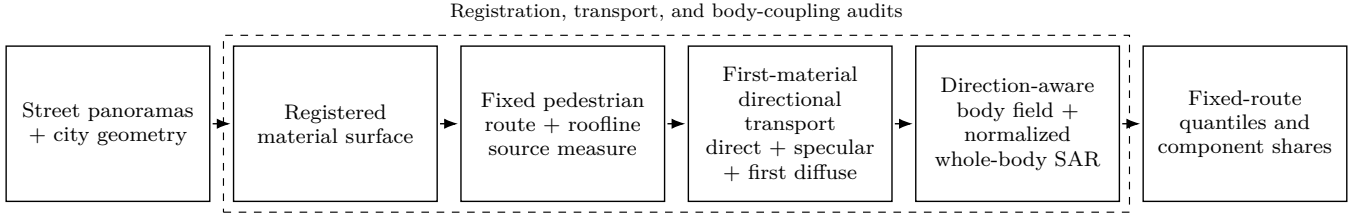


FIGURE 2. Flowchart of the computation and input checks. Aligned 360-degree street images and the city mesh give the material map. The fixed route and visible roofline give observation and transmitter positions. Transport retains direct, first-order specular, and first-diffuse terms. A hash-verified file list pins every input to the ten-site result.

square contribution of every roofline segment:

$$D_{\text{ref}}(\mathbf{x}) = \sum_i \frac{p_i}{r_i(\mathbf{x})^2}, \quad S_{\text{ref}}(\mathbf{x}) = \frac{A_{\text{crop}} D_{\text{ref}}(\mathbf{x})}{4\pi}. \quad (2)$$

No visibility test enters D_{ref} . All reported quantities are normalized per unit $\rho_A P_{\text{EIRP}}$. Multiplication by $\rho_A P_{\text{EIRP}}$ recovers physical units, but only for a deployment that follows the same roofline transmitter model. The calculation omits transmitters and interactions outside the crop.

The propagation result at each observation point has direct, one-reflection specular, and one-reflection diffuse parts. Let $\mathcal{D}$ contain the visible direct paths, and let $\mathcal{S}_1$ contain the accepted one-reflection specular paths. Their normalized powers are $\alpha_a = m_a^{(d)}/D_{\text{ref}}$ and $\beta_b = m_b^{(s)}/D_{\text{ref}}$. The diffuse estimate uses normalized angular-cell powers $\hat{\gamma}_q = \hat{m}_q^{(f)}/D_{\text{ref}}$. With $\delta_{\hat{\mathbf{k}}}$ denoting a unit point mass in physical arrival direction $\hat{\mathbf{k}}$, the directional distribution is

$$\begin{aligned} \mu_{\mathbf{x}} = & \sum_{a \in \mathcal{D}} \alpha_a \delta_{\hat{\mathbf{k}}_a^{(d)}} \\ & + \sum_{b \in \mathcal{S}_1} \beta_b \delta_{\hat{\mathbf{k}}_b^{(s)}} \\ & + \sum_{q=1}^Q \hat{\gamma}_q \delta_{\hat{\mathbf{k}}_q^{(f)}}, \quad Q = 4096. \end{aligned} \quad (3)$$

The first two sums retain the exact directions and powers of the direct and specular paths. They are not projected onto the angular grid. Only diffuse power is accumulated in the Q Fibonacci cells. Rays start at the receiver and travel outward to the first blocking surface, as shown in Fig. 3. Next-event estimation then tests visibility from that surface to every roofline segment [14]. The material model combines unpolarized Fresnel power with a Rayleigh roughness term. The remaining power enters a Lambertian diffuse term, and the components add incoherently. The sampled path ends after this diffuse reflection, and the calculation omits further specular reflections. Woody vegetation identified in the material map does not block rays because the city mesh has no canopy volume. The supplementary material gives the full laws and parameter values.

The arrival directions are carried through to the body. For outward body-surface normal $\hat{\mathbf{n}}(\mathbf{r})$, define $g(\mathbf{r}, \hat{\mathbf{k}}) = [\hat{\mathbf{n}}(\mathbf{r}) \cdot$

$(-\hat{\mathbf{k}})]_+$. The normalized absorbed power density is

$$\begin{aligned} \tilde{S}_{\text{ab}}(\mathbf{r}, \mathbf{x}) & \equiv \frac{S_{\text{ab}}(\mathbf{r}, \mathbf{x})}{\rho_A P_{\text{EIRP}}} \\ & = T_0 S_{\text{ref}}(\mathbf{x}) \left[\sum_{a \in \mathcal{D}} \alpha_a g(\mathbf{r}, \hat{\mathbf{k}}_a^{(d)}) \right. \\ & \quad + \sum_{b \in \mathcal{S}_1} \beta_b g(\mathbf{r}, \hat{\mathbf{k}}_b^{(s)}) \\ & \quad \left. + \sum_{q=1}^Q \hat{\gamma}_q g(\mathbf{r}, \hat{\mathbf{k}}_q^{(f)}) \right]. \end{aligned} \quad (4)$$

Here, T_0 is the normal-incidence power-transmission coefficient from the IT'IS tissue database at 15 GHz [15]. It approximates the absorption cross section for incoherent, unpolarized illumination. The function g sets the absorbed fraction to zero where the surface faces away from the incoming direction. The calculation applies this directional absorption to the 56,024-element Duke anatomical mesh [16]. Because all quantities are normalized by $\rho_A P_{\text{EIRP}}$, $\tilde{S}_{\text{ab}}$ is dimensionless. For the triangle area A_r at position $\mathbf{r}$ and body mass m_{body} , the normalized integrated quantities are

$$\begin{aligned} \tilde{P}_{\text{abs}}(\mathbf{x}) & \equiv \frac{P_{\text{abs}}(\mathbf{x})}{\rho_A P_{\text{EIRP}}} \\ & = \sum_{\mathbf{r}} A_r \tilde{S}_{\text{ab}}(\mathbf{r}, \mathbf{x}), \\ \widetilde{\text{SAR}}_{\text{wb}}(\mathbf{x}) & \equiv \frac{\text{SAR}_{\text{wb}}(\mathbf{x})}{\rho_A P_{\text{EIRP}}} \\ & = \frac{\tilde{P}_{\text{abs}}(\mathbf{x})}{m_{\text{body}}}. \end{aligned} \quad (5)$$

$\tilde{P}_{\text{abs}}$ has units m^2 , and $\widetilde{\text{SAR}}_{\text{wb}}$ has units $\text{m}^2 \text{kg}^{-1}$.

Each replica launches 200,000 independent and identically distributed primary rays per route point and accumulates first-diffuse power in 4,096 fixed Fibonacci output cells. Exact direct and specular paths bypass this grid. The output cells do not control the launch directions. The calculations use 16 replicas with seeds 7 through 22 and retain cumulative results after 4, 8, 12, and 16 replicas. Only the first-diffuse estimate varies between replicas. Scalar quantities and body fields are averaged across replicas before route statistics are computed. Route quantiles use linear interpolation over equally weighted route points, and the plotted empirical distributions use positions $(\text{rank} - 0.5)/n$. Multipath surplus is computed only

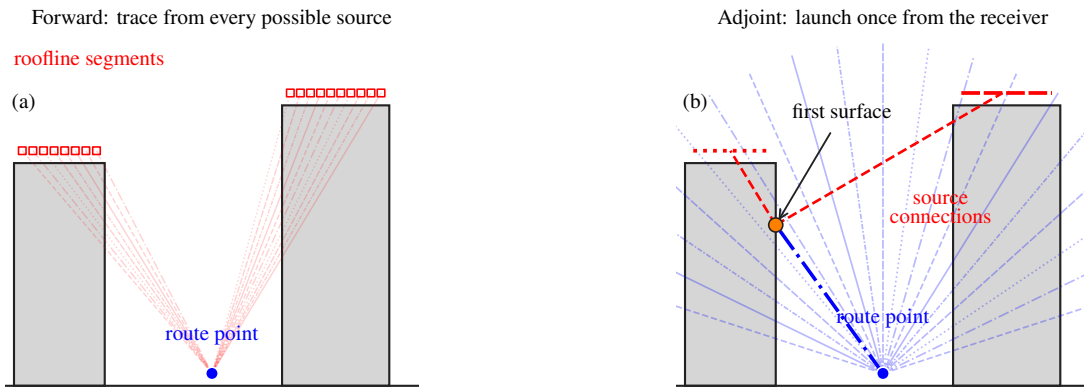


FIGURE 3. Forward and adjoint sampling for the first-diffuse term. (a) A forward calculation launches rays from every roofline segment toward the observation point. (b) The adjoint calculation launches rays once from the observation point. At the first blocking surface, connections to every visible roofline segment are tested.

where direct transfer is positive. A route point with zero direct transfer remains in the whole-body SAR distribution but has no finite surplus value.

III. RESULTS

Table 1 defines the ten fixed routes and their 163 observation points. Every site uses 15 GHz and a photogrammetric city mesh cropped to a 250 m radius. The Duke body model has 56,024 surface elements and a mass of 72.4 kg, and faces along the direction of travel. Each point has 64 independent replicas with seeds 7 through 70. Each replica uses 200,000 primary rays and 4,096 fixed Fibonacci output cells for the first-diffuse term. The resulting 10,432 body fields contain 2.086 billion primary rays. Calculation time for a prepared site ranges from 5.95 to 73.77 s on one A6000 GPU. These times exclude image acquisition, alignment, depth estimation, and material mapping because the full preparation time was not recorded.

A. VALIDATION

The calculation manifests list 42 files per site, and all 420 file hashes pass verification. The direct, exact order-1 specular, and first-diffuse fields sum to the stored total with a maximum absolute residual of $1.214 \times 10^{-17} \text{ m}^{-2}$ across all 10,432 fields. The GPU body-coupling result also agrees with the double-precision CPU reference to a maximum relative difference of 6.64×10^{-16} in the verified benchmark. These checks confirm the input files, addition of components, and agreement between CPU and GPU calculations. They do not externally validate the complete city model.

The direct and specular paths are computed from exact geometry, but the first-diffuse estimate depends on random ray sampling. The controlled comparison in Fig. 4 tests this stochastic component before the city results. The open-square scene has 27 sources, 6 receivers, 8 triangles, and one diffuse reflection. It has no specular reflection, refraction, or diffraction. Deterministic surface quadrature uses 2,097,152 samples. The adjoint estimate uses 50,000 primary rays for

each of four seeds. An independent Sionna RT forward calculation uses 50,000 samples per source for each of three seeds. The maximum difference between the adjoint estimate and quadrature is 0.0616 dB for the reflected term. A separate test of total transport gives a maximum adjoint-to-Sionna difference of 0.0344 dB. Fig. 4 plots the one-reflection transfer and its error relative to quadrature, not the total-transport test. The comparison checks first-diffuse normalization, visibility, inverse-square loss, and cosine terms in this one-reflection scene. It does not cover the image-derived material map, exact specular transport, or their combination in a city.

B. ROUTE EXPOSURE

Fig. 5 shows the fixed-route empirical distributions. Table 2 gives the central summaries and finite multipath surplus. Normalized whole-body SAR is reported in $\text{m}^2 \text{ kg}^{-1}$ per unit $\rho_A P_{\text{EIRP}}$. A physical whole-body SAR value therefore requires multiplication by the deployment's areal source density and EIRP. The figure includes all 163 observation points and marks the six with zero direct and zero order-1 specular transfer. The largest route median (Mexico City, $0.1291 \text{ m}^2 \text{ kg}^{-1}$) is 14.31 times the smallest (Milan, $0.009016 \text{ m}^2 \text{ kg}^{-1}$). Mexico City and Tokyo Hachiko have the widest spread along a route and the lowest tails.

The additive component shares of route-mean whole-body SAR are shown in the supplementary material. Of the 157 points with line of sight, direct transport is the largest contribution at 156. At one point in Brussels, exact order-1 specular transport is the largest. Mexico City points 0, 1, and 3 and Tokyo Hachiko points 13, 14, and 15 have zero direct and zero order-1 specular transport. First-diffuse transport is the only nonzero contribution at these six points. Across all 163 points, the pooled component medians are 78.056% direct, 20.231% exact order-1 specular, and 0.500% first diffuse. They do not sum to 100% because each component has a separate median over the 163 route points. The small first-diffuse median therefore does not describe the six fully shadowed points.

The path audit assigns each retained material interaction to

Controlled diffuse-reflection scene, 15 GHz, depth 1

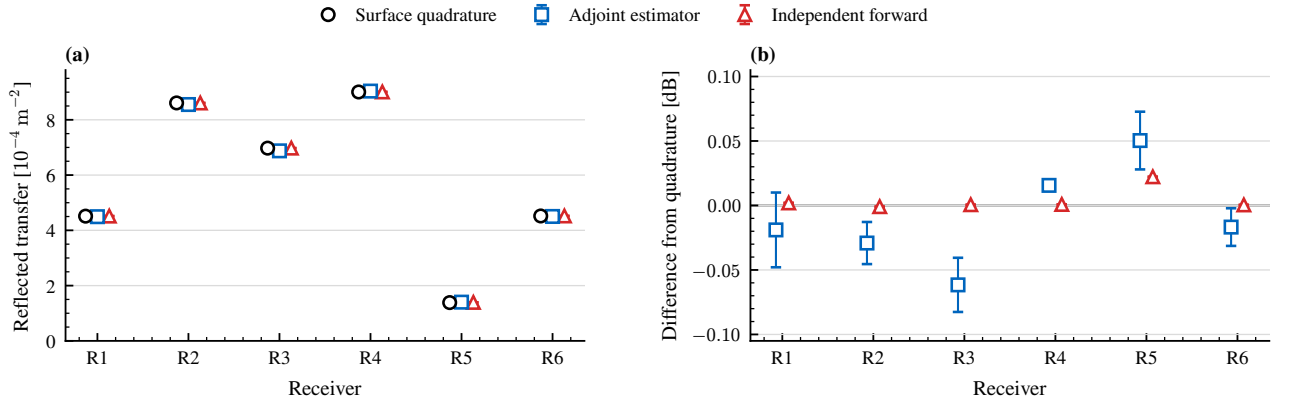


FIGURE 4. Controlled validation of the first-diffuse transfer. (a) Deterministic surface quadrature, the adjoint estimate, and the independent Sionna RT forward calculation at six receivers. (b) Signed error relative to quadrature. Error bars give the standard error across four adjoint seeds and three Sionna seeds.

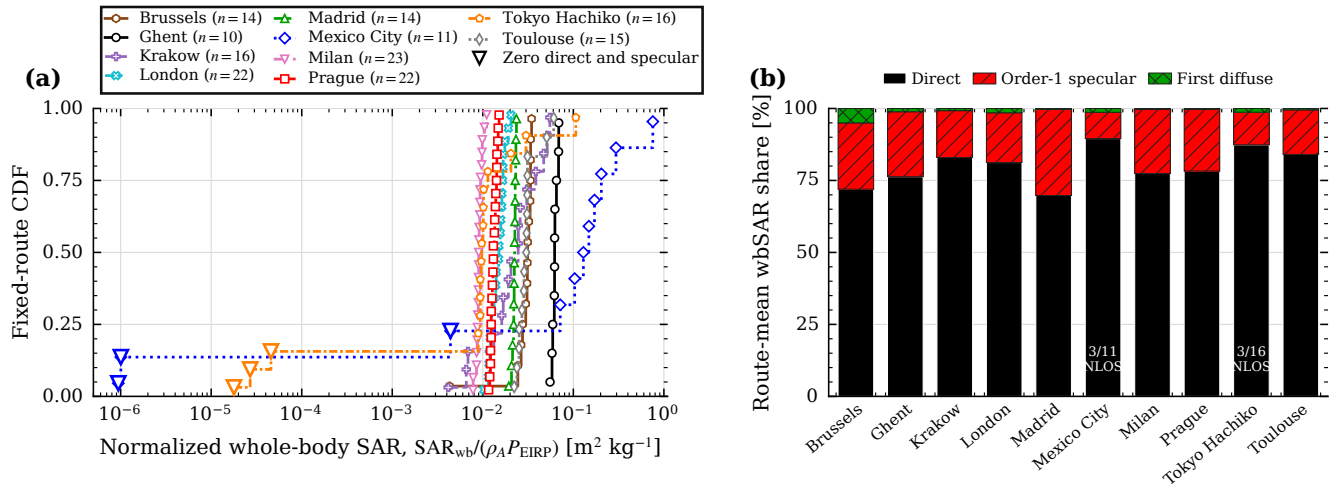


FIGURE 5. Normalized whole-body SAR on the ten fixed routes. (a) Empirical CDFs include all 163 route points. Hollow triangles mark the six points with zero direct and zero order-1 specular transfer. (b) Route-mean component shares of whole-body SAR. The routes are fixed case studies, not city or population samples.

TABLE 2. Fixed-route exposure summary. Whole-body SAR quantiles are normalized per unit $\rho_A P_{\text{EIRP}}$ and have units $\text{m}^2 \text{ kg}^{-1}$. Surplus is computed only at points with nonzero direct transfer. The last column is the maximum pointwise total-transfer change from 48 to 64 replicas.

Site	q_{10}	q_{50}	q_{90}	Median surplus [dB]	Zero direct and order-1 specular	Max total-transfer change [dB]
Brussels	0.025022	0.031187	0.033714	1.431	0	0.0003047
Ghent	0.057835	0.062047	0.068668	1.155	0	0.0001115
Krakow	0.006717	0.022520	0.049268	0.917	0	0.0002922
London	0.011746	0.015008	0.017444	0.901	0	0.0008481
Madrid	0.020913	0.022381	0.023170	1.552	0	0.0001015
Mexico City	9.92×10^{-7}	0.129061	0.295766	0.686	3	0.0138811
Milan	0.008529	0.009016	0.009861	1.136	0	0.0001842
Prague	0.012157	0.013074	0.014595	1.091	0	0.0001223
Tokyo Hachiko	3.66×10^{-5}	0.009674	0.025200	0.839	3	0.0055850
Toulouse	0.024312	0.030217	0.042915	0.835	0	0.0000651

an image-mapped or geometry-based surface. Pooled over the ten routes and 64 seeds, image-mapped surfaces account for 75.903% of the reflected SAR. The corresponding shares are 80.643% for exact order-1 specular transport and 13.337% for first-diffuse transport. The supplementary material gives the complete split by material source.

C. REPLICA CONVERGENCE

Between 48 and 64 replicas, the largest pointwise total-transfer change across all ten routes is 0.0139 dB at a shadowed point in Mexico City. The next largest is 0.00558 dB in Tokyo Hachiko. At the remaining eight routes, the maximum change stays below 0.001 dB. Route-median whole-body SAR is stable at all ten sites. In Mexico City, the single strongest diffuse replica at one shadowed point carries 5738 times the median replica contribution. This concentration produces the large lower-tail changes seen in Table 2 and is specific to the fixed routes. It does not represent city-sampling or route-selection uncertainty. The complete nested comparison is given in the supplementary material.

D. MATERIAL SENSITIVITY

A paired control for Madrid and Mexico City replaces all image-mapped materials with the default geometry-based materials while keeping the mesh, route, roofline model, body, seeds, sampling budget, and transport steps identical. Each reported change is $10 \log_{10}(x_{\text{image}}/x_{\text{geometry}})$. At Madrid the image-to-geometry changes in normalized whole-body SAR are +0.233, +0.249, and +0.269 dB for q_{10} , q_{50} , and q_{90} . At Mexico City the corresponding changes are +24.84, -0.158, and +0.104 dB. The large q_{10} change comes from the three fully shadowed points, where both totals are near zero and first-diffuse transport is the only nonzero contribution. The direct term is identical in every pair. At Madrid, the image-derived map changes the route-median specular component by +1.89 dB and the first-diffuse component by -12.43 dB, but the total median changes by only +0.249 dB. This control tests the complete material map, including its parameters and the treatment of woody vegetation as nonblocking. It does not measure material accuracy or isolate reflectance alone. The supplementary material gives pointwise and component-level comparisons.

IV. DISCUSSION

Route-median normalized whole-body SAR differs by a factor of 14.31 across the ten routes. The spread comes from differences in street width, building height, roofline visibility, surface materials, and canopy along each route. Each empirical distribution describes only that fixed route and its observation points. The selected routes do not define a city ranking.

The first-diffuse component is small at most points (0.500% pooled median), but it is the only nonzero contribution at the six fully shadowed points. Its small share where line of sight exists does not make it dispensable where buildings block all direct and specular paths. A transport

model that omits the diffuse term would assign zero exposure to those six positions.

The paired material control tests the effect of image-derived materials on the body results. The direct term is identical in both cases, so all observed changes come from the materials used by the reflected and diffuse terms. In Madrid, the specular and first-diffuse component changes have opposite signs, while the route-median normalized whole-body SAR changes by only 0.249 dB. The Mexico City route median changes by -0.158 dB. Its much larger lower-tail ratio comes from the three fully shadowed route points where both estimates are close to zero, not from a central effect. The paired cases differ in their assigned materials and in their treatment of woody canopy. The image-derived case treats identified canopy as transparent because the city mesh has no canopy volume. The comparison therefore measures sensitivity to both the material assignment and the vegetation rule together, and does not establish material accuracy on its own.

Several limits restrict what the results can say. The ten routes use one frequency (15 GHz), one body model facing along each walk, one roofline source model, and image-derived surface materials. Results are normalized per unit areal source density and EIRP. Scaling to a specific deployment is valid only if its transmitter positions follow the assumed roofline model. The transport model stops after one diffuse event and omits all later interactions and higher-order specular paths. The controlled comparison validates the first-diffuse component in a one-reflection scene. It does not validate the city calculations, which also use image-derived materials and exact specular transport. The 64 replicas quantify estimator randomness but not uncertainty in image-to-mesh alignment, geometry, material labels, route choice, transmitter placement, body shape, or body orientation. Ray calculation time ranges from 5.95 to 73.77 s per site on one A6000 GPU, but image acquisition and material mapping take longer and do not yet have a complete timing record.

The first priority is source calibration and end-to-end validation of the city calculation against outdoor field measurements of the directional field before body coupling. Second, higher specular orders and paths beyond the first diffuse event can be added through paired studies that report their change in whole-body SAR, variance, and computation time. Third, several routes at the same site can quantify route-selection variation within a single city. Other frequencies, body models, and orientations can then test the remaining range of validity. Each extension should also report what fraction of the surfaces reached by the modeled paths carries an image-derived material.

V. CONCLUSION

This study aligned 360-degree street images with a photogrammetric city mesh to map surface materials around pedestrian routes at 15 GHz. A roofline transmitter model and a directional absorption step then produced normalized whole-body SAR at ten urban locations, with all results per

unit $\rho_A P_{\text{EIRP}}$. In a controlled one-reflection scene, the first-diffuse estimate differed from deterministic quadrature by at most 0.0616 dB, and the maximum total-transport difference from an independent Sionna RT forward calculation was 0.0344 dB. Across 163 observation points on ten routes, route-median whole-body SAR differed by a factor of 14.31. Direct transport was largest at 156 of the 157 points with line of sight, while first-diffuse transport was the only nonzero contribution at the six fully shadowed points. The route medians were stable at 64 replicas, but the fully shadowed lower tails had larger estimator uncertainty. These results do not rank cities or predict deployed-network exposure.

The image-to-mesh material mapping took effect mainly through the specular component. A paired control at two sites showed route-median whole-body SAR changes of +0.249 and −0.158 dB when the image-derived materials were replaced by the geometry defaults. Ray calculation time ranges from 5.95 to 73.77 s per site on one A6000 GPU. The main open items are outdoor field validation against the full city model and a measured transmitter source distribution.

DATA AND CODE AVAILABILITY

The verified ten-site data, manifests, analysis scripts, and figure scripts are kept in the study repository. A stable public archive with a versioned digital object identifier will be deposited before publication. The 360-degree street images and commercial photogrammetric tiles are governed by their providers' terms and are not redistributed. Their identifiers and the alignment manifests are kept so the inputs can be reacquired where the licenses allow it.

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